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Deep Learning-Based Analysis of Mammographic Images for Breast Cancer Detection Using Transfer Learning

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ABSTRACT

Breast cancer ranks as one of the most perilous diseases, being the second most prevalent disease among women globally. This condition involves a rapid and uncontrolled proliferation of breast cells. Prominent symptoms of breast cancer include alterations in breast size and shape, discomfort, the presence of lumps, and nipple discharge. Primarily affecting women aged 50–60 years and older, this malignancy exhibits a substantial occurrence rate within the elderly demographic, accounting for over 80% of cases. Unfortunately, breast cancer claims the lives of numerous women annually. However, early detection and diagnosis offer a chance for a positive outcome, potentially reducing the fatality rate by 20% and enhancing patient survival rates. Past research endeavours have centred on training diverse models to detect breast cancer in its nascent stages. The objective of this research is to detect breast cancer at an early stage and facilitate medical fertility. In this work, we proposed an innovative transfer learning methodology for identifying breast cancer through mammograms. This approach not only aids medical professionals in automating diagnosis but also addresses potential human errors during analysis. By employing transfer learning, our technique empowers doctors to detect cancer in its incipient stages, substantially enhancing diagnostic accuracy. The convolutional neural network (CNN) architectures explored were Inception-v3, ResNet-50, VGG-16, SqueezeNet, and AlexNet. The dataset was collected from the local hospital of Lahore, comprising 900 mammogram images. After segmenting the mammogram images and eliminating noise, we conducted a comprehensive assessment of the CNN architectures. Remarkably, the AlexNet architecture outperformed its counterparts, achieving an outstanding accuracy of 96.7%. This achievement underscores the potency of our transfer learning model combined with distinct solver types.

1 | Introduction

Breast cancer is still one of the leading diseases among women for mortality and incidence in the world. Known as the malignancy that involves the breast cells, this disease can metastasise if the growth of the cells is not checked early. It is important to note that early symptom detection is essential, as it enhances the survival rates among patients and their partners, as well as improves the ways of their subsequent treatment. Over time, research into breast cancer has improved the diagnostic process and made technology more imperative to the diagnosis as well as the nature of the treatment of this disease, as it enhances the precision

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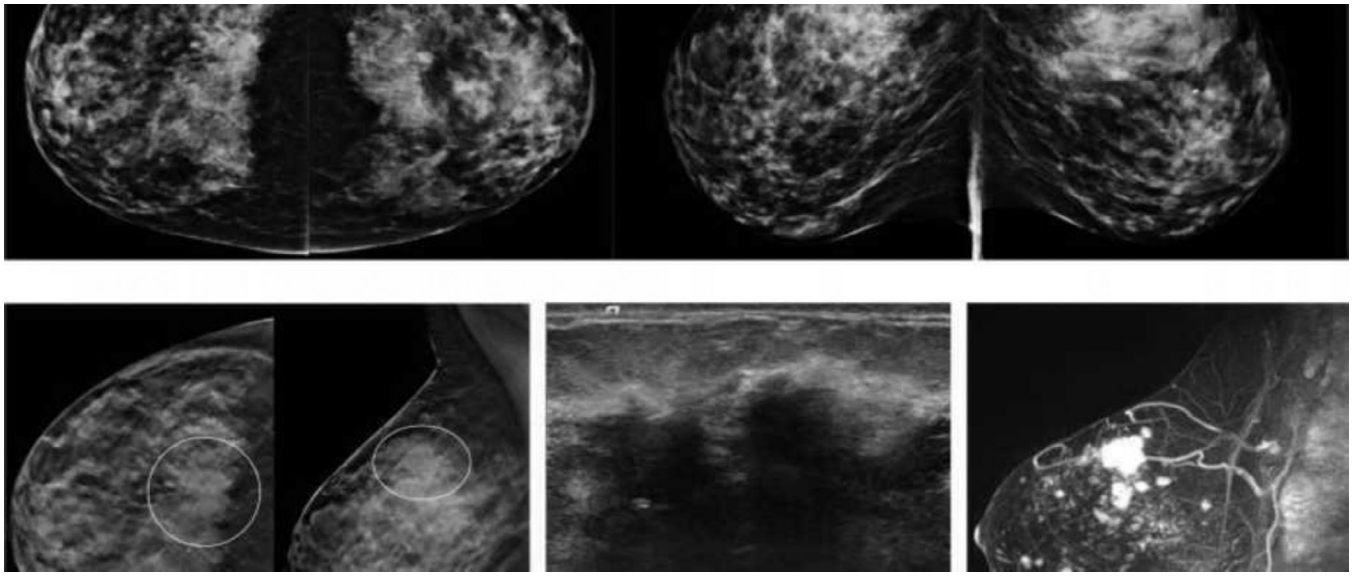


FIGURE 1 | Categorising tumours through various imaging modalities [10].

of health care [1]. One remarkable aspect of breast cancer is that it can be latent and without symptoms in the early stage, which makes prevention very crucial. Notably, breast cancers can be asymptomatic, meaning that they do not cause pain or the formation of lumps; hence, lack of traditional signs even at the beginning stage of the illness. In this regard, early diagnosis becomes invaluable in revealing all such hidden cases [2]. Using breast mammography, biopsy, tomosynthesis, breast ultrasound, computer-aided detection, PET, MRI, and X-ray forms a vital part of the tools used in the detection of breast cancer [3]. Mammography as an initial diagnostic procedure can contribute to a decrease in mortality, especially breast cancer-related deaths. Mammography can be considered an essential primary diagnostic method for tumours and significantly contributes to reducing the mortality rates of breast cancer [4]. Besides tumour identification, mammography provides important information on skin lesions within breast tissues [5]. To achieve highly accurate results, radiologists are expected to study mammograms and go deeper into scrutinising any abnormal tissues within the breast. To improve the effectiveness of disease detection, it is advisable to review the results of mammography more than once in order to have a detailed look at everything. One advancement in this field is Full-Field Digital Mammography (FFDM), in which there are digital scans of the breasts, and these are stored in the system [6]. In practice, a mammogram device centres the breast and use low-dose X-rays to produce images. However, these X-rays may not give comprehensive reports of the abnormal tissues [7]. In an attempt to overcome this shortcoming, the breast is gently compressed between the machine plates, making it easier to distinguish the pathologic tissue from the surrounding anatomy. Medical images are one of the most investigated areas in the context of the image processing domain. Not only do these images help in the acquisition of features, but they also help in experimental surgeries and treatment control. Specific to the above is breast tumour diagnosis, a critical task requiring expert involvement, as the disease has complex differences in diagnosis. Many methodologies of employing different software solutions for breast tumour diagnosis have been put forward to make this process more efficient and convenient for patients. Currently, various screening methods can be used in the identification of breast cancer at its initial stages. These include thermography, ultrasound, mammography, clinical biopsy, magnetic resonance imaging (MR), fine needle aspiration cytology, and tommodensitometry [8]. While these methods are utilised for disease detection, some of the techniques are constrained in certain ways, making some of them impossible, or patients reject them due to the painful procedures and exposure to radiation. As an available modality to most patients, ultrasound imaging brings some limitations such as noise distortions that hinder the delineation of abnormal areas. On the other hand, effective MRI is very costly and unavailable in areas of low income [9]. Figure 1 illustrates the categorising tumours through various imaging modalities [10].

Automated and stabilised breast treatment using mammogram data is a significant sub-discipline in computational methods that seeks to identify abnormalities in breast tissue from medical image data [10]. After diagnosis, patients may seek lumpectomy surgeons for invasive surgical procedures to examine and evaluate the mammary tissue zone [11]. Breast tumours fall into two types: benign and malignant. Since they are malignant, tumours invading the breast and surrounding tissues exhibit characteristic signs. On the other hand, benign tumours do not invade sections of the breast and do not possess the potential to harm. These distinctions are discerned in mammograms that distinguish between healthy, malignant, and normal cases [12]. Figure 2 shows the malignant, non-malignant, and healthy breast imaging results [13].

Breast tumours, specifically malignant ones, require a multidisciplinary approach—computerised scans, mammography screenings, MRI scanning, positron emission tomography scans, thermal imaging, ultrasound imaging, and clinical biopsy—to diagnose and manage both benign and malignant breast diseases [13]. However, the method that has been found effective and recommended by many specialists is mammography screening. It is a cheap, noninvasive procedure that uses low-energy X-rays to capture the images. However, X-ray imaging is considered a powerful tool for the visualisation of abnormal tissues with

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FIGURE 2 | Malignant, non-malignant, and healthy breast imaging results [13].

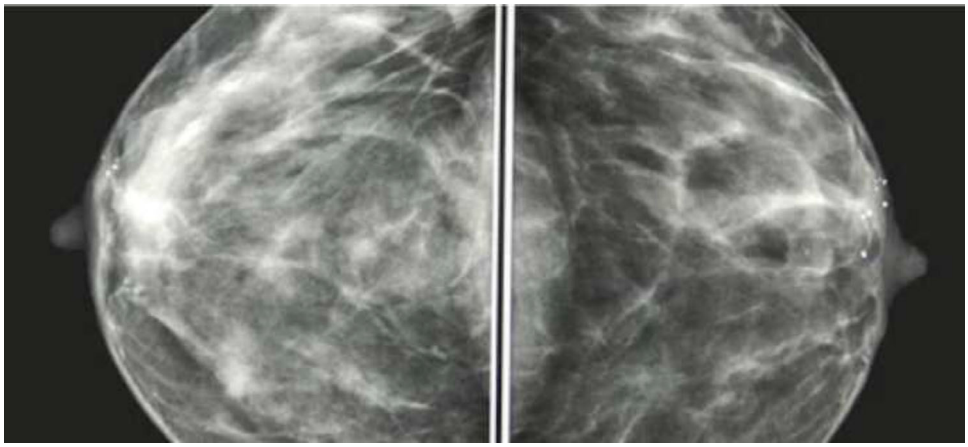


FIGURE 3 | Mammographic screening [13].

low doses. These rays are well-regulated to reduce impacts on other vital body organs. Modern medicine has led to 3D mammography [14], which does not use radiation as the previous methods did. This, though, is still relatively expensive, but it is more economical than other methods used in diagnosis, such as MRI. Mammography is cost-effective and hence possible for middle-income bracket regions [10]. However, some limitations exist, such as the fact that occasional radiologists may miss diseases during mammograms. In response to this concern, a detection system has been developed to detect interest regions in mammograms [15]. This system is useful to the radiologist as the second observer that aids the diagnostic process. CAD helps radiologists detect any abnormality in lesions in digital mammograms and minimise false negatives. The study also shows that different types of cancer at a preliminary stage can effectively be detected by CAD [9, 16]. Thus, X-ray imaging presents itself as a quickly developing, accurate method for breast tumour assessment. Breast tumour mammograms are very beneficial in hospitals, as they provide information concerning several comparisons of heterogeneous breast tissues. Since various forms of cancer exist, and considering that each type of mammogram is restricted to a particular view, doctors acquire craniocaudal (CC) and oblique mediolateral (MLO) views of each mammogram. Two primary features are used for tumour detection, namely, masses and microcalcification signatures [17]. Figure 3 shows the mammographic screening [13].

Masses represent space-occupying lesions, predominantly malignant, and micro-calcifications represent small deposits of calcium within the breast tissue, which are seen as white or grey opacities on mammography. In general, masses are often considered benign, and micro-calcifications are benign. However, using general estimates made by radiologists can lead to errors because of tissue differences, noise, and indistinct edges [18]. To these challenges, different approaches, based on manually extracted features, including history elimination, segmentation, region of interest extraction, and classification, have been proposed. Feature extraction methods include graphical, textural, and mathematical properties [10]. Different types of classification techniques, including K-Nearest Neighbours, support vector machines, decision trees, and Naive Bayes, have been used. These classifiers produce favourable results and are central in overcoming the problems associated with manual breast tumour sorting. Among image processing techniques, thresholding is essential in erasing signatures and other figurative marks with the mammograms' context information. River classification and other forms of computerised image assessment rely on clustering-based methods to eliminate breast cancer tissues [19]. Therefore,

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X-ray imaging has become a rapid and reliable method of examining breast tumours. CAD-enhanced mammography provides information about various breast conditions affecting the breast, including cancer. Although there are issues with the methodologies, future improvements in automated methods will lead to improvements in the accuracy of early breast cancer diagnosis. A synergistic application of deep learning and the proficient AdaBoost algorithm was implemented, harnessing datasets sourced from online repositories, resulting in an accuracy of 81%. Another innovative approach employed an edge-assisted deep learning model for breast cancer screening, leveraging the DMR-IR dataset to achieve an impressive 93.8% accuracy.

Recent advances in deep learning have revolutionised medical image analysis, offering new opportunities to enhance the accuracy and efficiency of breast cancer detection systems. Many cutting-edge deep learning approaches address key challenges in mammographic interpretation: attention mechanisms, generative adversarial networks (GANs), graph neural networks (GNNs), explainable AI (XAI) techniques, and ensemble methods. These approaches collectively address critical challenges in breast cancer screening, from data limitations and interpretation variability to the need for clinical transparency [20]. Attention mechanisms enable models to prioritise diagnostically relevant regions in mammograms, with architectures like the self-attentive CNN achieving 94.92% accuracy in classification tasks. GANs address data scarcity through synthetic image generation, with contextual in-filling GANs successfully synthesising lesions on healthy mammograms. GNNs model complex relationships between mammographic regions through node-edge representations, demonstrating the potential for capturing spatial dependencies. XAI techniques bridge the gap between model predictions and clinical interpretability, providing visual heat maps of suspicious regions. Ensemble methods combine multiple models to enhance detection reliability, with four-model averaging strategies achieving 0.98 AUC on FFDM. Their integration promises to develop next-generation computer-aided detection systems that improve early diagnosis while maintaining compatibility with existing clinical workflows. Despite these advancements, there remains room for refinement, especially concerning early-stage accuracy. In this paper, a total of 3600 mammographic images of 900 patients taken from a local hospital situated in District Lahore have been gathered, and different pre-trained convolution models have been applied to these images. On this novel dataset, pre-trained models of Inception-v3, ResNet-50, VGG-16, SqueezeNet, and AlexNet have been applied to the CNN. AlexNet has achieved an accuracy of 96.7. The study [21] takes advantage of an openly available GitHub dataset. This allows academics to evaluate the classifiers' performance and distinguish between breast cancer and its absence. The researchers discovered that ten-fold cross-validation outperforms five-fold cross-validation. Fold cross-validation helps to improve results. It also considers the consequences of different split per cents and finds that 66% of the total distribution is the optimal value. This case highlights the need to use suitable validation methods while machine learning is classifying breast tumours. Important for the evolution of cancer diagnosis prediction models and computational medicine, the results show that the J48 decision tree approach performs better than the others. The main goal of this work [22] is to develop a 3D imaging and artificial intelligence monitoring-based early breast cancer detection method. The mammography images undergo preprocessing to enhance their overall quality after collection. One uses a multistage selective convolution filter (MSCF) to lower noise. We additionally use an enhanced contrast limited adaptive histogram (ECLAH) to boost contrast. An improved canny operator (ICO) is lastly used to simplify edge detection. dense se-net, or dense squeeze excitation network, is applied for segmentation. By altering preprocessed images, a technique called digital breast tomosynthesis (DBT) produces a three-dimensional look. Object recognition in integrated two- and three-dimensional images is accomplished with the You Only Look Once v7 (YOLO v7) algorithm. After classifying the images into four groups: mild, medium, severe, and unaffected women, the semi-supervised convolutional neural network (SS CNN). This research [23] starts with the background of breast cancer, the effect of the disease, the significance of mammography screening, and the classification methods with machine learning techniques in medical image analysis (Section 1). Section 2 presents the literature review that provides an overview of previous approaches to breast cancer detection. This section mainly includes artificial intelligence, CNNs, transfer learning, YOLOv5, and AlexNet. In Section 3, the proposed methodology is outlined, detailing the five-phase process. Section 4 presents the results and discussions, including the dataset description, preprocessing, training, and validation results, with comparisons to existing models. Section 5 provides a comparison of the proposed method with the previous studies to analyse the efficiency of the model. Last, Section 6 closes the research by providing an overview of the results and recommendations for future studies on the enhancement of breast cancer detection using deep learning. This work presents a consistent and accurate machine vision approach based on symmetry CT scan images for lung tuberculosis detection. The atypical development of tuberculosis (TB) might cause uneven or limited effects on the lungs, either in one area or another. Thus, from both normal and tuberculosis-infected lung CT scans, regions of interest (ROIs) were extracted following completion of the necessary pre-processing steps: selection/cropping, greyscale image conversion, and filtering. After obtaining statistical texture features, we optimised 30 features using F (Fisher) + PA (probability of error + average correlation), plus MI. Out of these, we decided on the top six features to work on at the last minute.

1.1 | Motivation of the Study

One of the main causes of cancer mortality, breast cancer affects women differently. Early diagnosis and greater survival rates have a clear relationship. Conventional diagnostic techniques, such as mammography, are prone to human errors and can be improved with the aid of artificial intelligence. Particularly, CNNs, deep learning models, have shown amazing performance in image recognition challenges. By means of these models for mammography image analysis, radiologists could find hitherto missed anomalies. Constructing deep learning models from the ground up calls for access to vast datasets and substantial computational resources. Using pre-trained models with feature knowledge from large image datasets—such as ResNet, VGG, or Inception—transfer learning enables researchers to achieve remarkable accuracy with limited medical data by optimising these models with mammography data. They thus have great chances for success. Reducing the number of erroneous diagnoses

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TABLE 1 | Comparison of existing research methodologies.

Study	Dataset used	Model/ technique	Input views	Accuracy/ AUC	Strengths	Limitations
Khan et al. (2021) [21]	MIAS, DDSM	SVM, k-NN with handcrafted features	Single view	AUC ≈ 0.78–0.85	Simple, interpretable	Poor generalisation, sensitive to dense tissue
Ribli et al. (2018) [22]	INbreast	ResNet-50 (DL)	Single view	Accuracy: 91.2%	High sensitivity, end-to-end learning	Lacks multi-view correlation
Shen et al. (2019) [23]	NYU Breast Screening Dataset	Self-Attentive CNN	Dual-view (CC, MLO)	Accuracy: 94.92%	Attention-based, interpretable heatmaps	Requires large data
Hsieh et al. (2022) [24]	CSAW-CC	GNN	CC only	AUC: 0.98	Models spatial dependencies	Computationally intensive
Zhuang et al. (2023) [25]	VinDr-Mammo	Vision Transformer with Multi-Head Attention	CC + MLO	Accuracy: 96.5%	High accuracy, transformer strength	Less interpretable
Dembrower et al. (2020) [26]	Swedish National Registry (463k patients)	Deep Learning (AI-Triage System)	All standard views	Cancer detection by 17.6%	Real-world validation	Requires a clinical integration pipeline

is the aim since it can result in undetectable cancer or needless treatments. Deep learning models are good at spotting trends in images that the human eye would overlook. By means of the system to acquire a second opinion, radiologists can enhance their diagnostic confidence and efficiency.

The motivation behind this study is to facilitate medical professionals in the early detection of breast cancer using mammographic images through computer-aided solutions, as well as human beings, to save time and start their medication as early as possible. The primary value of this research stems from its access to real-time clinical data obtained from a local hospital. Using actual time datasets provides both better algorithm performance and reduced error rates. Early breast cancer detection through this research will lower the mortality rate among breast cancer patients. Medical experts can use mammogram images to diagnose breast cancer accurately. Through this research, doctors gain the ability to analyse mammogram images for efficient diagnosis. The study aims to enhance accuracy and results by applying various CNN architectures to specific datasets. Table 1 illustrates the limitations of the work.

The rest of the paper is organised as follows: Section 2 presents a literature review, discussing various studies that utilise AI, ML, and DL in cancer research. Section 3 illustrates the methodology, providing a detailed description of the dataset, preprocessing, training, and testing. Section 4 presents the results and disscussion and Section 5 concludes the study.

2 | Literature Review

In the study by [24], the authors utilised mammography for accurate breast cancer diagnosis. The strategy that they adopted followed a sequential plan. Firstly, they overcome the image noise problem by applying the median filter for noise removal. After that, image segmentation was performed with the help of a CNN technique to separate the tumour regions [25]. To enhance efficiency, some features were removed while others were extracted using an appropriate method. An SVM classifier was used for feature training to differentiate between abnormally cancerous tissues and healthy tissues. The Grasshopper Optimisation Algorithm was applied to both image segmentation and feature selection problems. The proposed technique was compared to 10 other methods using the Digital Database for breast cancer datasets, and when compared with different techniques, it achieved 92% improvement in accuracy. The authors [6] proposed the OMLTS-DLCN technique for breast cancer diagnosis and classification. The proposed approach incorporated many sections.

The suggested adaptive fuzzy-based feature fusion (AFFF) was used to remove noise, increasing image transparency. In breast screening specifically, the OKMT-SGO approach is used for detecting malignant areas. Moreover, feature extraction was performed using the Capsule Network (Caps Net Model). Finally, the Backpropagation Neural Network (BPNN) technique was developed for classification purposes. That is why, taking into consideration the Mini-MIAS and DDSM datasets, the effectiveness and classification outcomes of this approach were confirmed. From this proposed approach, an average accuracy of 89% was overall achieved; the classification results were further compared with other methods using this significant improvement. In this proposed study, the various methodologies utilised for breast cancer diagnosis and classification showcase advancements in noise reduction, feature extraction, segmentation, and classification techniques to improve diagnostic accuracy.

An automated detection of breast cancer using mammography is enabled through the application of artificial intelligence.

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The proposed study introduced a DL (Deep Learning) model specifically designed for the detection of breast cancer in digital mammograms with changing frequencies. Execution and achievement of this proposed technique presented more accurate results than previous research work. The dataset consisted of 1,501 images of full-field digital mammograms that were processed, and the resulting images were 302 from February 2007 to May 2015. There are two convolutional neural network (CNN) models well-trained for identifying cancerous lesions within these merged images. Subsequently, the performance of these models was evaluated for 301 extracted images from 284 subjective images. To provide context, the study's results were contrasted against a meta-analysis of 12 previous research methods. A correlation between breast density and the efficacy of malignancy detection is revealed in this research. When the ability to detect malignancies decreased, the breast density increased. Importantly, the deep learning technique showcased heightened effectiveness in diagnosis and screening tests of breast cancer within FFDM of varying frequencies, with particular improvement observed in cases of breasts with low parenchyma density. Achieved an accuracy rate of 95% for detecting breast cancer from the proposed method [26].

The deep learning-assisted AdaBoost algorithm applied a novel approach for the early-stage diagnosis and detection of breast cancer tissues. This study leveraged the power of both deep learning and ensemble techniques to improve predictive accuracy—the proposed methodology employed an AdaBoost technique to construct an ensemble classifier for the final prediction function. Through a comprehensive evaluation, it was demonstrated that this novel algorithm exhibited superior predictive competence. The deep learning classifiers outperformed other types of classifiers, demonstrating heightened performance and yielding optimal results [27]. The proposed DLA-EABA model shows high potential for achieving fast generalisation and enhancing predictive functions in a completely autonomous mode, thanks to its neural networks. A special example of this approach is the deep learning CNN-based method called DLA-EABA, which was described in [28]. This particular method is quite useful in improving system performance, especially in situations where the deep learning technique is applied. Further, the process is used to specify attributes in datasets, making it consistent with the machine learning paradigm.

The utilisation of the DLA-EABA technique proved to improve the level of accuracy of mass detection of breast cancer, thus helping reduce the mortality of breast cancer patients. For better detection of breast cancer and guidance of future studies, this proposed study focuses on assembling techniques with deep learning. A leading contributor to female mortality is breast tumours, which stand as a significant health concern, particularly among women. Enhanced survival rates for breast cancer patients have been achieved through advanced diagnostics and treatment, facilitated by technological advancements. This paper introduced a novel technique for classifying breast thermograms into malignant and non-malignant categories, leveraging an ensemble transfer learning method. Through the utilisation of transfer learning, the computational time required for training the classification model was significantly reduced, enhancing overall efficiency. The results of this study are compared with those of previous studies. This proposed technique attained an exceptional accuracy rate of 96.25% and an F1 score of 0.96. The proposed ensemble transfer learning approach has the potential to precisely classify breast thermograms, ultimately assisting in enhancing diagnostic abilities and more effective surgery [29].

The breast tumours were detected at an early stage utilising transfer learning (TL). The objective of this work is to attain more efficient and accurate results from deep learning by using mammogram datasets compared to conventional machine learning methods. The proposed approach involved using the pre-trained AlexNet deep learning model through transfer learning, adapting it to address the issue with minimal resource usage [3]. The methodology consisted of two main phases: preprocessing and training, followed by testing. A dataset from the Kaggle repository was employed, comprising ultrasonic and histopathology images for the early diagnosis of breast tumours. The three categories of cancer are successfully classified in this approach—benign, malignant, and normal—which helps in the decision of treatment, whether medication is more needed. At present, artificial intelligence techniques include machine learning and deep learning technologies, which are used in different areas of work, specifically in medical image analysis. The machine learning methods used to detect breast tumours at an early stage involve screening mammograms with MRI [30]. The proposed method consisted of two layers—preprocessing and testing. MRI scan images were transformed and fed into a pre-trained AlexNet50 algorithm. The proposed trained algorithm was then applied to test images for prediction. In case unhealthy tissues were detected, the patient was advised to consult a doctor; otherwise, there's no need for consultation [28]. The artificial intelligence was incorporated into computer-aided decision-making (CAD) for the analysis of breast ultrasound images. Two datasets of breast ultrasound images, containing malignant and benign cases, were used for CAD analysis [28]. Various algorithms were compared using CAD for tumour detection, with Viola-Jones and YOLOv3 being trained for the task. The models were trained with radiomics signatures, and their accuracy, sensitivity, and specificity were evaluated. The proposed YOLOv3 technique outperformed the Viola-Jones algorithm, achieving better accuracy and outcomes.

In this research, we develop a framework for the detection of breast cancer using mammographic images. The dataset obtained for this research work was from a local hospital situated in Pakistan. This dataset has four labels, including BIRAD 1 to IV. In this work, to classify these images, five pre-trained models, including AlexNet, VGG-16, Inception v3, ResNet-50, and SqueezeNet, are used. The contribution of this research work includes:

- Mammographic images obtained from the local hospital are gathered.
- CLAHE has been used for preprocessing the mammographic images.
- Five pretrained CNN models are used for feature extraction and detection of cancer stages in mammographic images.
- Performance matrix, such as accuracy, F1 score, sensitivity, and specificity, is used to evaluate the performance of the proposed method.

Table 2 provides the limitations of previous work. In [37], a deep learning algorithm-based intelligent analysis model of the

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TABLE 2 | Limitations of previous work.

References	Dataset and size	Approach	Applications	Accuracy
[31]	CBIS-DDSM (2178 mammograms)	CNN	Breast cancer identification and classification	81.02%
[32]	OPTIMAM (3524 mammograms)	Transfer Learning	Breast mass detection	88%
[33]	CBIS-DDSM (2478 mammograms)	Transfer learning-based method	Mammography for breast cancer detection	94.13%
[34]	Public datasets (10,509 numeric datasets)	A combination of machine learning and deep learning techniques	Breast cancer diagnosis	93.6%
[6]	MIAS and DDSM (322 mammograms)	Machine learning algorithms	Mammogram image-based breast cancer detection	89%
[35]	CBIS-DDSM (2478 mammograms)	Deep learning model for improved detection	Breast cancer detection	87%
[26]	BI-RADS (301 mammograms)	EfficientNet-B5 and DenseNet-169 architectures	Automated breast cancer identification	81%
[27]	Wiki cancer imaging archive (1000 cell images)	DLA-EABA technique	Early-stage breast cancer detection	81%
[24]	DDSM and MIAS (100,000 numeric data)	CNN-based approach	Breast cancer diagnosis	92%
[36]	BreakHis (7909 cell images)	Inception-v3 and ResNet-based models	Histopathology-based breast cancer detection	87%
[37]	SERS spectra antibiotic mixture datasets	CNN, NN-EN, SHAP	antibiotics in aqueous solution	98.68%
[38]	knee osteoarthritis MRI	VGG11, DenseNet121, ResNet18, 3D-CNN	knee osteoarthritis detection	87.5%
[39]	mini-MIAS mammography, INBreast, and CBIS-DDSM	AlexNet, GoogLeNet, VGG16, ResNet-50	Breast cancer detection	88.59%

SERS spectrum was proposed in this study to quickly identify which components of the antibiotic are in the mixture and quantitatively determine the ratio of the components. Developed a water environment system with three antibiotic residues of ciprofloxacin, doxycycline, and levofloxacin. To enable both qualitative and quantitative analysis of the SERS spectra antibiotic mixture datasets, we designed a computational framework that combines a CNN with an adaptive image segmentation algorithm—a network (CNN) and a non-negative elastic network (NN-EN) approach. The experimental data indicate that the CNN model has a recognition accuracy of 98.68%, and the explanation analysis of Shapley Additive exPlanations (SHAP) suggests that our model can specifically target the characteristic peak distribution. Conversely, the NN-EN model is capable of correctly measuring the ratio of each component in the mixture. The combination of the SERS method with the aid of the CNN and NN-EN model has a high potential for distinguishing and quantitating antibiotic remnants in aquatic environments quickly and with high precision.

In the work by [38], the possibility of using popular CNN architectures (ResNet, DenseNet, VGG, and AlexNet) to differentiate between knees with and without osteoarthritis (OA) is explored. They demonstrated the potential of 3D CNNs, comprising 13 various architectures, in diagnosing knee osteoarthritis using 3D convolutional layers. We employed transfer learning to convert 2D pre-trained weights into 3D weights, serving as the starting weights for training 3D models. Comparison of the models was done based on their performance. Comparison was done in terms of the performance measures: balanced accuracy, precision, F1 score, and area under the receiver operating characteristic (AUC) curve. They recommended that, indeed, transfer learning improved the performance of the models, particularly ResNet and DenseNet models. Transfer learning-based models gave encouraging results, with ResNet34 getting the highest overall accuracy of 0.875 and an F1 score of 0.871. It also demonstrated the performance of shallow networks compared to deeper neural networks, showing superiority as evidenced by ResNet18, DenseNet121, and VGG-11, which have AUCs of 0.945, 0.914, and 0.928, respectively. This promotes the use of clinical diagnostic support of knee osteoarthritis with 3D CNN, even in limited conditions.

The [39] proposes a framework of mammogram preprocessing based on an optimised deep learning ensemble technique to identify microcalcifications. The process begins with artefact elimination through Otsu segmentation and morphological processing, followed by image resizing and adaptive median filtering. It then involves the development of a deep convolutional neural network (D-CNN) using transfer learning with the ResNet-50

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model. The hyperparameters are tuned, and an ensemble of models (AlexNet, GoogLeNet, VGG16 and ResNet-50) is developed to localise microcalcifications. The ensemble model achieves an average confidence of 0.9305 for microcalcification and 0.8859 for normal cases, indicating high accuracy, precision, recall, F1-score, and AUC. This performance surpasses that of other models, confirming the model's reliability.

3 | Methodology

The approach presented for the diagnosis and detection of breast cancer using mammograms is a comprehensive approach for the classification of the multiclass breast cancer images. The literature survey of the preceding studies has explored the applicability of ML techniques for BIRADS classification using mammogram datasets. In the proposed work, a mammogram images dataset was collected from the Local Hospital of Lahore. This work aims to use the CNN approach to detect breast tumours from mammogram images. CNN structures used in this proposed approach are Xception, Inception-v4, ResNet, and AlexNet, which are distinct from each other. These structures were trained by using the transfer learning technique. The real-time dataset consisted of mammogram images collected from 900 patients at a local hospital in Lahore. In the preprocessing phase, these mammogram images were filtered to remove noise and enhance their quality. The processed mammogram images were then fragmented and classified to eliminate all abnormal tissues while also completely separating normal tissues using mammography. Architectures of CNN are then used to perform classification on the mammogram images, using machine learning filters and pixels to look for objects in the images. This process involves partitioning pixels using different signatures to identify image features. The collected dataset is used to train CNN models of various architectures, enabling them to generalise the learnt information to other unknown data. By experimenting with the Xception, Inception-v4, and ResNet architectures, their performance is carefully evaluated, and the results are contrasted. The outcome is a model that has the highest accuracy level. This is a five-phase approach with the overall goal of identifying breast cancer through mammography. This process is not only technologically advanced but also ethically correct, due to collaboration with the local hospital in Lahore and following the proper techniques for data management.

3.1 | Dataset Acquisition

This data collection takes place at the Department of Radiology of a hospital in Lahore, after receiving ethical clearance through the institutional research committee. The dataset is made from 1,350 mammogram images of 920 patients, along with the physicians' diagnostic reports. These records are divided into four classes: Class I, Class II, Class III, and Class IV, and help in the classification of tumour severity. The dataset is based on the cases of breast cancer treated between the years 2018 and 2020 to give context to the study's analytical requirements. Table 3 shows the dataset details.

Demographics of the dataset are described in Figures 4 and 5.

TABLE 3 | Dataset description.

Sr. No.	Dataset	Total patients	No. of records	Type of data
1	Mammogram	900	1350	Images

3.1.1 | Significance of the Dataset

The data was gathered in a local hospital in Pakistan, representing South Asian women as a demographic group that is frequently under-represented in publicly available data. This further makes our data more applicable to the regional screening programmes and models specific to this population. The data consists of recent mammograms captured in the present clinical settings and using modern imaging devices, so the images represent the current clinical state. In contrast, some publicly available datasets use outdated imaging technologies. The hospital board-certified radiologists reviewed and labelled all the images based on the standards of BI-RADS. This guarantees annotation of high quality and clinical validation. The data contains a broader range of breast densities and pathological variations and cases of both rural and urban populations to improve the robustness of the model on the various patient profiles. The images it contains are closely associated with histopathology reports, which will serve as a better source of ground truth than the data used in the present study, which is based purely on radiological evaluation.

The dataset comprises 1130 mammography images collected from a premier tertiary care facility in Lahore, Pakistan, which serves a diverse patient population from across the country, with lots of patients' daily visits from various ethnic groups and geographic regions. The dataset encompasses a wide range of demographic profiles, lesion types (e.g., fibroadenoma, mammary carcinoma, and benign proliferative lesions), and imaging characteristics, captured using standardised, state-of-the-art mammography equipment. This diversity, coupled with standardised imaging protocols using state-of-the-art mammography equipment, ensures that the dataset captures a broad spectrum of demographic and clinical variations, enhancing its representativeness of Pakistan's population and potentially other South Asian regions with similar breast cancer epidemiology. Figures 4 and 5 show the description of the dataset's composition, including patient demographics, lesion types, and imaging characteristics. We acknowledge the limitations of relying on a single-centre dataset in the limitations section and propose future validation on external datasets, such as CBIS-DDSM or INbreast, to confirm model robustness across diverse imaging conditions and populations.

3.2 | Data Processing

This phase involves extensive preprocessing of the mammogram images that have been obtained from the database. It is important because it brings sharpness to images and removes noise that has no role in image analysis. After this, the data is divided and sorted to get ready for further analysis. All these preprocessing steps improve the quality and credibility of the final set, improving the framework and foundation of data mining and analysis in

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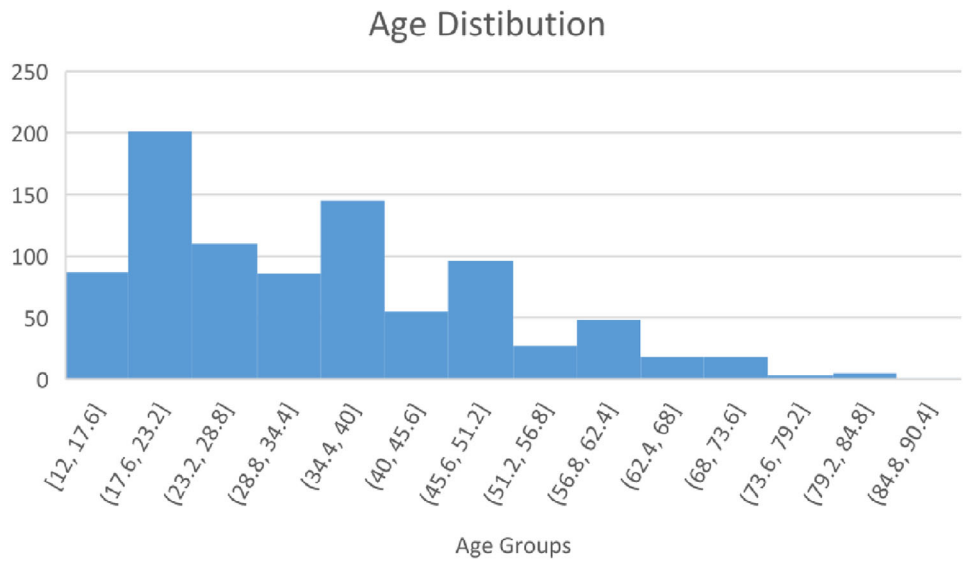


FIGURE 4 | Age distributions of patients.

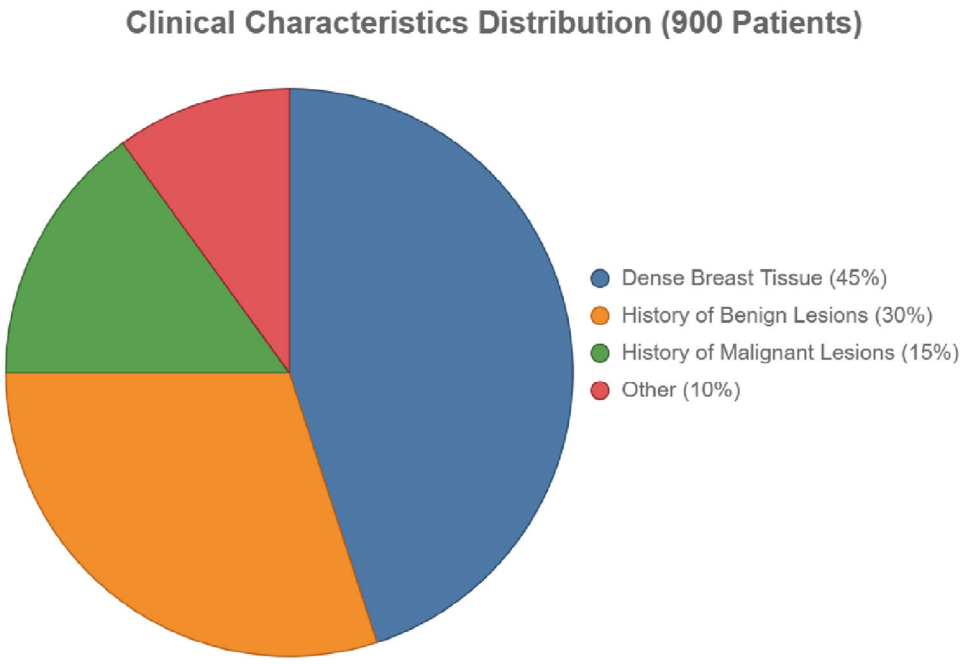


FIGURE 5 | Clinical characteristics distribution.

subsequent levels. The stringency of these processes ensures dataset quality to fit the standard needed for insight generation.

3.2.1 | Noise Removal

The mammogram images have noise, which is removed using a Gaussian filter of a 3x3 kernel and standard deviation (sigma) of 1.5. This kind of smoothing is quite good at reducing high-frequency noise, but not at the expense of important edge information, the detection of which is key to abnormality detection. Also, the Contrast Limited Adaptive Histogram Equalisation (CLAHE) is used to boost local contrast; the clip limit is 2.0, and the tile grid size is 8x8 to make tiny details visible in dense breast tissue. Image normalisation: All images are scaled to a common size of 512×512 pixels to provide uniformity in the whole dataset. The pixel intensities are then rescaled to the range of [0, 1] to enable them to be compatible with the machine learning algorithms. Figure 6 shows the mammograms before and after noise removal.

A kernel size of three was chosen for computational efficiency and localised smoothing. The value of 1.5 was optimised to a standard deviation that effectively reduces Gaussian noise without excessive edge blurring, as evidenced by a 12.3 dB increase in PSNR compared to unfiltered images. Various kernel sizes (5×5, 7×7) and sigmas (0.5, 1.0, 2.0) were tried, and 5x5

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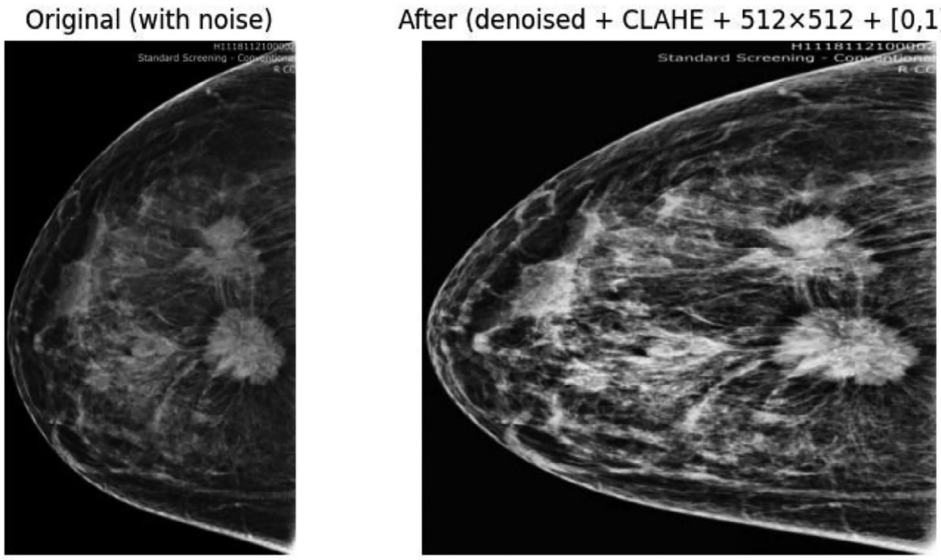


FIGURE 6 | Mammograms before and after noise removal.

(sigma = 1.5) had a PSNR improvement of only 10.8 dB and a loss of 15 per cent of edge detection accuracy, whereas 2.0 sigma caused over-smoothing (20 per cent edge blur). These findings informed our choice of the 3*3 kernel with sigma = 1.5 to be used in the preprocessing of the data before applying CNN.

In Phase 4, the processed mammogram dataset is used in developing and evaluating models for diagnosing breast cancer correctly. Various machine learning algorithms are used to analyse their accuracy and potential for future applications. Such an approach of validation makes it possible to compare the models to test their performance on new data other than the training set. The purpose of this phase is to compare results from different models and select the optimal diagnostic model based on the identification of strengths and weaknesses of each model. Figure 7 shows the proposed methodology.

The last step involves a detailed assessment of the results obtained from the constructed models. Here, the performance of the models developed in this study is compared to the performance from other studies to demonstrate improvements made by the proposed method. This comparative analysis confirms the effectiveness of the proposed model, thus estimating how well it would solve the research problem. This phase concludes with a presentation of the study's contributions, focusing on the practical advancements achieved in breast cancer diagnosis through improved image analysis.

3.2.2 | CNN Architectures

This proposed technique involves transfer learning combined with CNN methods for the classification of breast cancer. Transfer learning requires adjusting the layers of the CNN model to fit the problem as defined in this research. The attributes of the input layer are standardised in MATLAB to achieve a fixed size and dimension of the mammogram image in the dataset. Here, the input layer adjustment and data preprocessing are described.

Inception-v3 uses multi-scale convolutional filters in the inception module layers to enable the detection of lesions of a large variety and complexity, which is critical when detecting fine lesions in dense breast tissue. The vanishing gradient problem is solved by ResNet-50, through its residual connections, enabling the deeper layers to learn and utilise the information provided by the lower layers and improve its ability to learn hierarchical features that are important to discriminate between BIRADS I-IV stages. VGG-16 has a deep, sequential convolutional architecture that is very effective in capturing fine-grained texture information and is essential to microcalcification and other fine-grained abnormality detection. The computational efficiency of SqueezeNet balances accuracy and the cost of computation, making it suitable for a resource-limited clinical setting. All these architectures collectively provide an in-depth analysis, complexity, and efficiency in transfer learning of breast cancer detection, which ensures robust performance in mammographic patterns.

Image Resizing for Consistency: Each image I is resized to dimensions $W \times H$ for consistent input. Given mammogram images I_n in the data set shown in Equation (1)

$$I'_n = \text{resize}\ (I_n\ ,\ \text{W},\ \text{H}) \tag{1}$$

Where I'_n Show the resized mammograms to uniform the size.

The resizing function ensures all images are the same size, allowing the network to treat each image uniformly without considering their sizes. It helps in achieving a standard parameterisation of the shapes throughout the images.

Feature Extraction Layer: The model initially adopts pre-trained weights $W_{i,j}^{TL}$ During the training phase, these weights are fine-tuned to adapt relevant features for breast cancer classification. Where i denotes the number of layers, and j denotes the neurones.

Fully Connected Layer: In this layer, the extracted features f is converted into a prediction of each class. The layer is designed to

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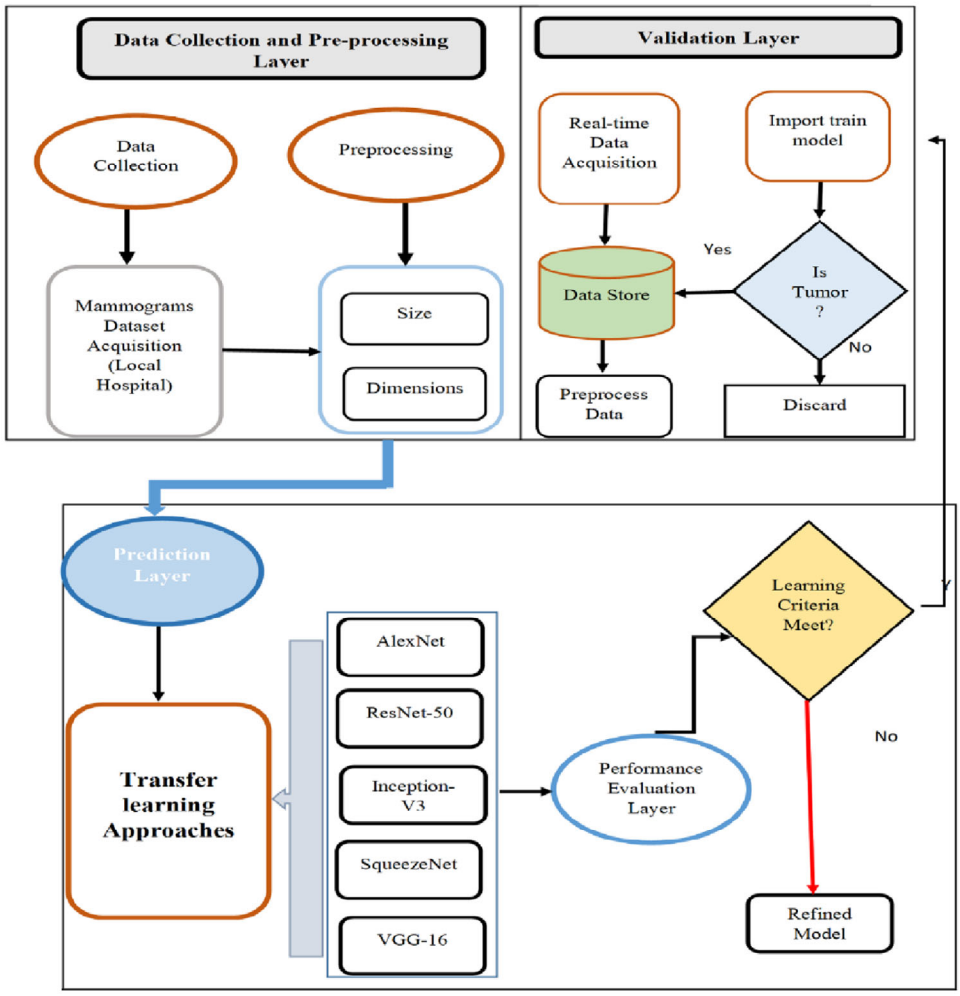


FIGURE 7 | Proposed methodology.

classify five classes of breast cancer, each of which represents a different stage of the disease. The operation of FCL is shown in Equation (2)

$$FCL(f) = W_{FCL} \cdot f + b_{FCL} \tag{2}$$

FCL(f) provides estimates from the learnt features utilising weights W_{FCL} and b_{FCL} Show the bias vector of the layer that matches all the features with each class in the correct manner.

The SoftMax layer is used to derive probability objectives for classification classes. For an image I'_n, let the output of the FCL for each class c_i be z_i.The SoftMax layer computes the probability $P(C_i \mid I'_n)$ as:

$$P(C_i|I'_n) = \frac{e^{zi}}{\sum_{j=1}^{c} e^{zi}} \tag{3}$$

Where e^{zi} Exponential of the input score for class i. C is the total number of classes.

The proposed study makes it possible to perform a comparison of the probability results obtained by the various classes of breast cancer.

CNN even mimicked the training to enhance both the efficiency and the optimisation of the learning rate of the architecture. To overcome these difficulties, a new fuzzy technique is developed for learning rate selection. A fuzzy technique assigns an initial learning rate $\eta_0 \mathbb{E} = (0, 1)$, dynamically adjusted epochs t using a decay function for stability and optimisation.

$$\eta_0 \mathbb{E} = \eta_0 . e^{-\lambda t} \tag{4}$$

Where λ is the decay factor balancing the rate of convergence, the learning rate affects the convergence of the algorithm, and the optimisation is controlled by setting an initial rate (IR) between 0 and 1 during training and classification. The best learning rate refers to the degree of flexibility in training time while at the same time seeking to achieve the highest performance. For model training, the train network function is utilised, which encompasses data training, layer training, and model training options. During training, the cross-entropy loss L measures the discrepancy between predicted probabilities and true labels:

$$\mathrm{L} = -\sum_{i=1}^{c} yi.logP(C_i \mid I'_n \tag{5}$$

Where y_i is the true label for class i. The function trains the model, which can be done only once in this case. The next step in the

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model's analysis utilises the analyse network function to evaluate the effectiveness of the pre-calculated data. The given dataset is divided into training and testing. In each class, 80% of the data is used for model training, while the remaining 20% is used for testing assessment. During the training process, the plot option in the training options is enabled to display such things as accuracy, loss, learning rate, and validation progress. For evaluation on the test set, accuracy A is calculated as:

$$A = \frac{\text{Number of correct prediction}}{\text{Total number of Prediction}} \tag{6}$$

The plotted mechanism proposed in the article enables the tracking of the training progress in real time and can be used to assess the model's performance and accomplishments.

Therefore, this involves data pre-processing, using transfer learning on common CNN architectures, and using strategies such as using SoftMax layers and adjusting learning rates for better classification of breast cancer. The dataset mentioned above was split into training and testing sets to evaluate the efficacy of the proposed model based on the visualisation and analysis of the pertinent parameters.

3.3 | CNN Model

A CNN may have convolution layers through which actual convolutions occur. These layers may then be followed with pooling or normalisation, or any other layers, depending on the architecture of the implementing network. CNNs have different architectures; however, there is always a convolutional layer in charge of feature extraction.

In the convolutional layer, a kernel or filter matrix is used to scan the input image to extract features. The convolution operation can be defined mathematically as the sum of the element-by-element product of the image and the filter. While dot products are suitable for this purpose, other modifications are better or worse than others depending on the situation.

In Equation (7), the output of the convolution is computed using the kernel and input image:

$$\text{net}(i,j) = (\xi * \varphi)[i,j] = \sum_m \sum_n \xi[m,n]\varphi[i-m, j-n] \tag{7}$$

In Equation (7), where $net(i,\ j)$ is the output at position (i,j) is the input image, and φ is the kernel or filter.

A CNN's first element is the convolution layer, which is used for feature extraction within the image. Suppose we have an image of size 7×7 having RGB channels, a kernel of 5×5, and the scan over the image is done with a stride of 1. The output feature map's dimensions can be calculated as follows (Equation 7):

$$\text{out} = \frac{(W - F + 2P)}{S} + 1 \tag{8}$$

Where W is the size of 2D, F is the filter size, P is the padding, and S is the Stride. For the example with a 7×7 image, a 3×3 kernel, and a stride of 1, the output size is:

$$\text{out} = \frac{(7 - 3 + 2(0))}{1} + 1 = 5 \tag{9}$$

Thus, the output feature map will be of the size 5×5. After that, to introduce non-linearity, the ReLU activation function is applied to the output. Although ReLU is standard in CNNs, it has a problem in that it is not differentiable at zero, which complicates backpropagation. The function is defined as:

$$y = \max(0, x) \tag{10}$$

And its derivative is:

$$\frac{dx}{dy} ReLU(x) = \begin{cases} 1, & if\ \ x > 0 \\ 0, & otherwise \end{cases} \tag{11}$$

The ReLU is plotted to understand its operational mode and where the output is zero for negative value inputs and the activation input for positive value inputs. After the set of convolution layers, the layers of max-pooling are applied to diminish the spatial size of an image, thus diminishing both its height and width. For instance, a max-pooling layer with a 2×2 filter size and 2 strides is used. The final fully connected layer is succeeded by a SoftMax layer that normalises the output into a probability distribution to give class outcomes.

3.4 | Mathematical Model of CNN Loss

In a CNN, it is possible to quantify the error between the forecast and the given values via the loss function. We also have Y, which is the target values from the convolutional layer, and y, which stands for the predictions. The loss can be computed as the mean and used to update the affinity through backpropagation.

To reduce this loss, we estimate the derivative of the loss with respect to weights and biases. The most commonly used loss function for classification tasks is cross-entropy loss, defined by:

$$L = -\sum_{i=1}^{c} Y_i \log(y_i) \tag{12}$$

Where c is the number of classes, the SoftMax function transforms the logits (raw network outputs) into probabilities, as shown in Equation (12):

$$y_i = \frac{e^{Z_i}}{\sum_{k=1}^{n} e^{Z_k}} \tag{13}$$

Where Z represents the logits.

The logits are obtained by computing the weighted sum of inputs defined in Equation (14):

$$Z_l = \sum_{j=1}^{n_{\text{out}}} W_{jl} X_j \tag{14}$$

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Where Z_l shows the logit value for output neurones and n_{out} total number of input feeds for output. $W_{jl}X_j$ shows that weights and activation function

To update the weights, we calculate the derivatives of the loss function with respect to the logits, as shown in Equation (14):

$$\frac{\partial L}{\partial W_{jl}} = \sum_{j=1}^{n_{\text{out}}} \sum_{l=1}^{c} \left(\frac{\partial L}{\partial Z_l} \times \frac{\partial Z_l}{\partial W_{jl}} \right) \tag{15}$$

The derivative can be broken down into two cases based on whether the predicted class is the same as the true class.

In Case 1, where $i = \text{l}$ the derivative is:

$$\frac{\partial y_i}{\partial Z_l} = y_i \left(1 - y_i\right) \tag{16}$$

In Case 1, where $i \neq 1$ the derivative is:

$$\frac{\partial y_i}{\partial Z_l} = -y_i y_l \tag{17}$$

These derivatives are summarised as:

$$\frac{\partial y_i}{\partial Z_l} = \begin{cases} y_i \left(1 - y_i\right), & \text{if } i = l \\ -y_i y_l, & \text{if } i \neq l \end{cases} \tag{18}$$

Finally, the derivative of the loss with respect to the weights W_{jl} is computed as:

$$\frac{\partial L}{\partial W_{jl}} = \sum_{j=1}^{n_{\text{out}}} \sum_{l=1}^{c} \left(y_l - Y_l\right) x_j \tag{19}$$

This value is used in gradient descent to update the weights and biases, thereby optimising the network's performance.

4 | Result and Discussion

In this research, the experiments were conducted on a Z Book G16 with 32 GB RAM, a 1 TB hard drive, a 3.2 GHz processor, and a 6GB dedicated GPU. The primary software used in performing the experiments is MATLAB 2021. The dataset of every class is divided into two partitions: 80% for training and 20% for testing.

This study utilised five distinct CNN algorithms in our transfer learning model: VGG-16, Inception-v3, SqueezeNet, AlexNet, and ResNet. Each of these techniques has its unique architecture and characteristics, which contribute to their effectiveness in detecting breast tumours from mammogram images. This proposed VGG-16 training algorithm involves 100 epochs, and each epoch consists of eight steps. In using the Adam solver, accuracy is stabilised at 74 epochs by using the VGG-16 model with the Adam solver. The details of these parameters are shown in Table 4. Training is terminated after reaching the maximum number of epochs, which is determined by the system. Figure 8 shows the training of the VGG-16 model using the Adam solver.

The choice of AlexNet, Inception-v3, ResNet-50, VGG-16, and SqueezeNet was based on their unique architectural benefits in the classification of mammographic images in the detection of breast cancer. AlexNet, with its comparatively shallow eight-layer architecture, was selected because it was computationally efficient and had converged quickly on a small dataset, which makes it appropriate to detect the characteristic features like masses and microcalcification. Inception-v3 uses stacked inception modules with multi-scale filters, which allow capturing of various-sized lesions and complex patterns simultaneously, which is a critical requirement in detecting subtle abnormalities in dense breast tissue. The deep, residual-connected ResNet-50 overcomes the issue of vanishing gradients and enables the extraction of complex hierarchical features that are important in differentiating the BIRADS I-IV stages. The deep, sequential 16-layer model VGG-16 is particularly good at finding fine-grained textural features that are critical to the detection of microcalcifications and other subtle mammographic abnormalities. SqueezeNet, which is optimised with efficiency in mind, provides a light architecture that trades off accuracy and computing cost, making it a possible candidate in resource-limited clinical environments. In combination, these architectures offer a comprehensive assessment of various levels of depth, complexity, and computational efficiency, enabling the detection of breast cancer in mammographic images.

TABLE 4 | Models' hyperparameters.

Pre-trained models	Iterations	Batch size	Epochs
VGG-16	800	10	100
Inception-v3	800	10	100
SqueezeNet	800	10	100
ResNet-50	800	10	100
AlexNet	800	10	100

The learning rate of the proposed model in this work is based on a fixed schedule of the constant value of 0.02. The training is conducted in a single mode using one CPU, which demonstrates the efficiency of the proposed model in terms of the number of CPU calls. For training, the process can perform a maximum of 800 iterations, and it is planned to comprise eight iterations per epoch. The model processes the images in 10 chunks in the initial training phase. The model, in its initial form, is only 30% accurate. As more iterations are done in training, the result is increased accuracy of the model being built. When the training is at 400 iterations, which is 50 epochs, the model's accuracy is at about 80%. Improved performance in this model illustrates the learning from the data process that is going on. The training process finishes 800 iterations, which means that the model makes 100 epochs, and the maximum achieved accuracy is 99.22%. From this proposed model, the images have been learnt and categorised appropriately as planned. In the case of VGG 16 trained on the Adam solver, it was observed that the loss rate was over 5000 in the first iteration itself. This high loss rate in the beginning is okay because the model is starting to learn right from scratch. Each time model training occurs, the loss rate decreases, indicating that the model is adjusting its parameters to minimise inaccuracy. In general, this proposed training model indicates a clear positive trend in the sense that the accuracy is improving while the loss rate is declining as the number of iterations rises. Employing Adam Solver in this paper improves the VGG-16 model's learning

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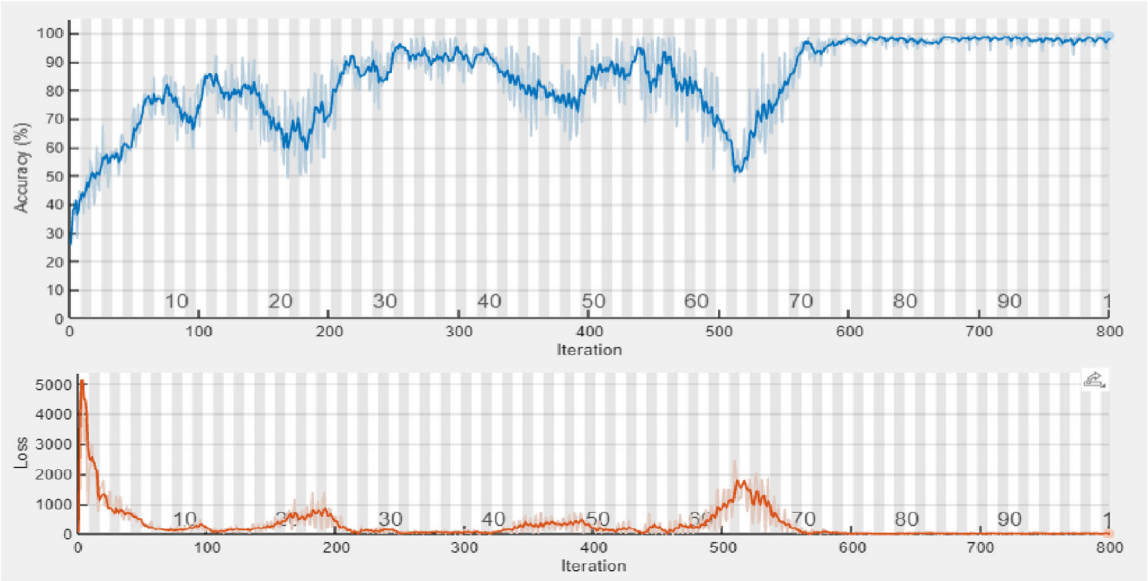


FIGURE 8 | Training of the VGG-16 model using the Adam solver.

TABLE 5 | Performance analysis of validation results.

Model	Solvers		
	Adam	**RMSprop**	**SGDM**
VGG-16	93.8%	77.6%	92.9%
SqueezeNet	93.5%	81.7%	93.8%
ResNet-50	74.3%	65.2%	92.6%
AlexNet	**96.7%**	94.5%	95.6%
Inception-v3	95.0%	72.9%	95.0%

capability to improve the accuracy and reliability of the model for diagnosing breast cancer from mammograms.

To test our proposed transfer learning framework, five CNN architectures are utilised: VGG-16, Inception-v3, SqueezeNet, AlexNet, and ResNet. The validation process of these five algorithms is conducted utilising three varied solver types: Adam, RMSprop, and SGDM. In this study, the AlexNet transfer learning model is used. It is validated using three different solvers: Adam, RMSprop, and SGDM. The validation results of the AlexNet algorithm are shown in Table 5.

The particular hyperparameters of each solver to guarantee replicability are Adam with learning rate 0.001, beta1 0.9, beta2 0.999, and epsilon 10 -8; RMSprop with learning rate 0.001, decay rate 0.9, and epsilon 10 -8; and SGDM with learning rate 0.01 and momentum 0.9. These solvers were chosen over others (e.g., Adagrad, Adadelta) because of their demonstrated flexibility in medical image classification applications, with the Adam adaptive learning rate being appropriate in the case of mammography features with changing gradients, RMSprop momentum stabilising training on noisy data, and SGDM momentum accelerating convergence in deep networks and the diverse lesion patterns (e.g., masses, calcifications) in our dataset.

The accuracies in the performance of the mammography image classification differ due to the differences in the model architecture, number of parameters, transfer learning efficiency, and compatibility with the dataset. The simpler structure of AlexNet with large filters probably extracts coarse mammography features well, and the multi-scale Inception modules of Inception-v3 are more suitable for recognising a wide variety of patterns, but not as ideally as AlexNet. VGG-16 and SqueezeNet are good alternative options but could be constrained by high parameters (VGG-16) or low complexity (SqueezeNet), resulting in overfitting on fine details or underfitting, respectively. The low performance of ResNet-50 may be attributed to its excessive depth, resulting in overfitting on a potentially small or imbalanced training set, or to its ineffective fine-tuning of pretrained weights, such that its more complex residual learning is not as well-adapted to mammography patterns as compared with the proposed architecture, AlexNet.

To make the proposed CNN models more interpretable and applicable in clinical practice, we did an interpretability analysis using Gradient-weighted Class Activation Mapping (Grad-CAM) to understand how the five models make their decisions, including AlexNet, VGG-16, SqueezeNet, ResNet-50, and Inception-v3. The test set was used to perform this analysis and consisted of 226 mammography images within a collection of 900 patients in one of the leading tertiary hospitals of Lahore, Pakistan. Grad-CAM produces heatmaps that show regions of the mammogram to which the model pays attention when making a classification, giving clear insight into how it makes decisions. In AlexNet, the best model with an accuracy of 96.7 per cent, the suspicious areas were well localised in malignant cases (e.g., BIRADS III-IV) as revealed in the heatmaps, and this was coincident with the presence of atypical cells. Inception-v3 and SqueezeNet performed well on the localisation (87.6% and 86.3% correct ROI localisation, respectively) but performed less precisely in dense breast tissue, whereas VGG-16 and ResNet-50 performed less consistently (82.7% and 79.2%), sometimes highlighting non-diagnostic areas. Representative Grad-CAM visualisations

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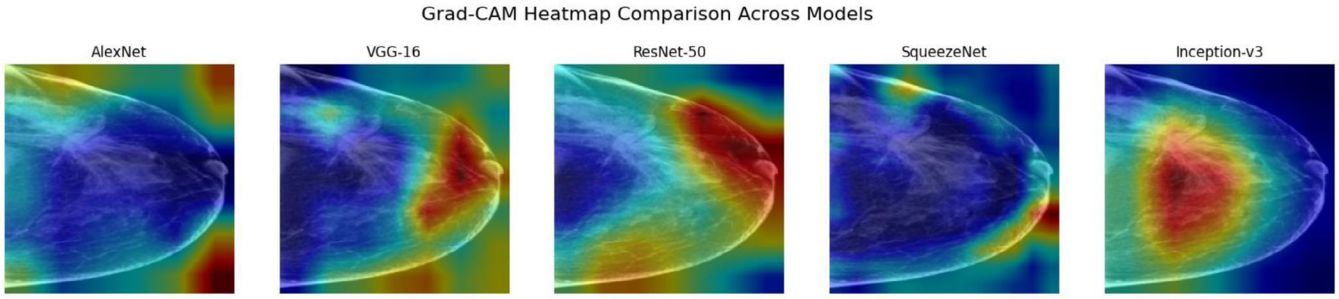


FIGURE 9 | Grad cam heatmap.

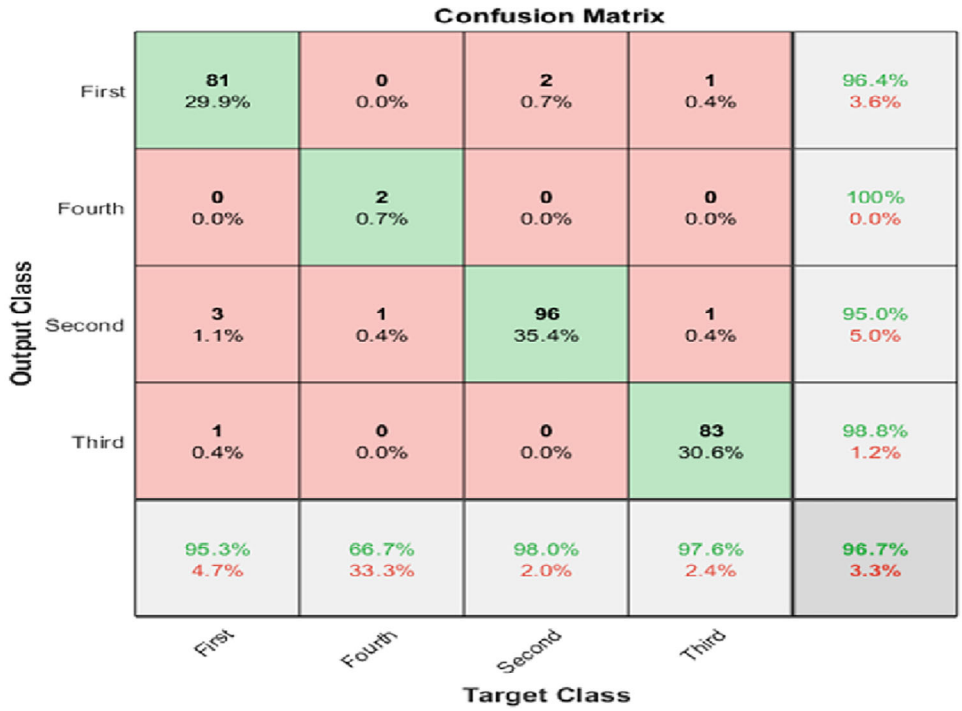


FIGURE 10 | Confusion matrix.

are provided in Figure 9, overlaid on mammography images to demonstrate that the models focused on the key diagnostic features. Also summarises the interpretability performance, quantifying the proportion of cases where heatmaps correctly localised ROIs as validated by expert annotations.

4.1 | Confusion Matrix

The confusion matrix of the proposed transfer learning model with the AlexNet algorithm and the Adam solver is as follows. The dataset is categorised into four classes: first stage cancer BIRADS I, second stage cancer BIRADS II, third stage cancer BIRADS III, and fourth stage cancer BIRADS IV. The horizontal axis is the target classes, and the vertical axis is the predicted classes. For the first class, the model correctly recognises 81 images while incorrectly classifying three images as belonging to the third and fourth classes, out of 84 images. This leads to 0.964 accuracy and 0.036 misclassification rate. The second class yields 101 images, with 96 images properly classified and 5 incorrectly classified into the first, third, and fourth classes; hence, an accuracy of 95.0% and a misclassification of 5.0%. With the third class, there is 83/84 = 98.8% accuracy and a misclassification rate of 1.2%. Finally, it is noted that all images belonging to the fourth class are classified without any error, giving a classification accuracy of 100% and a misclassification of 0.0%. In total, the AlexNet model with the Adam solver has a confusion matrix that suggests a total accuracy of 96.7% and a total misclassification of 3.3%. Figure 10 shows the confusion matrix.

The confusion was between BIRADS II (benign) and BIRADS III (probably benign) categories, with 4% (9 out of 226 images) misclassifications, which were mainly due to similar characteristics, namely subtle calcifications or dense tissue patterns that resemble malignancy at an early stage. Also, 2% (5 images) of BIRADS IV (suspicious) were mislabelled as BIRADS III, probably due to the presence of atypical pleomorphic cells that the model did not sufficiently capture, since it focuses on hyperchromatic nuclei. These errors can be attributed to the variability of the lesion sizes (e.g., 1 × 1 cm to 12 × 9 × 7 cm) and imaging conditions of the

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TABLE 6 | Evaluation results.

Pretrain model	Accuracy	Precision (weighted)	Recall (weighted)	F1-score (weighted)
VGG-16	93.8%	94%	93.7%	93.7%
SqueezeNet	93.5%	92.3%	93.5%	93.5%
ResNet-50	74.3%	71.9%	74.2%	74.2%
Inception-v3	95.0%	94.0%	95.0%	95.0%
AlexNet	**96.7%**	**96.8%**	**96.7%**	**96.7%**

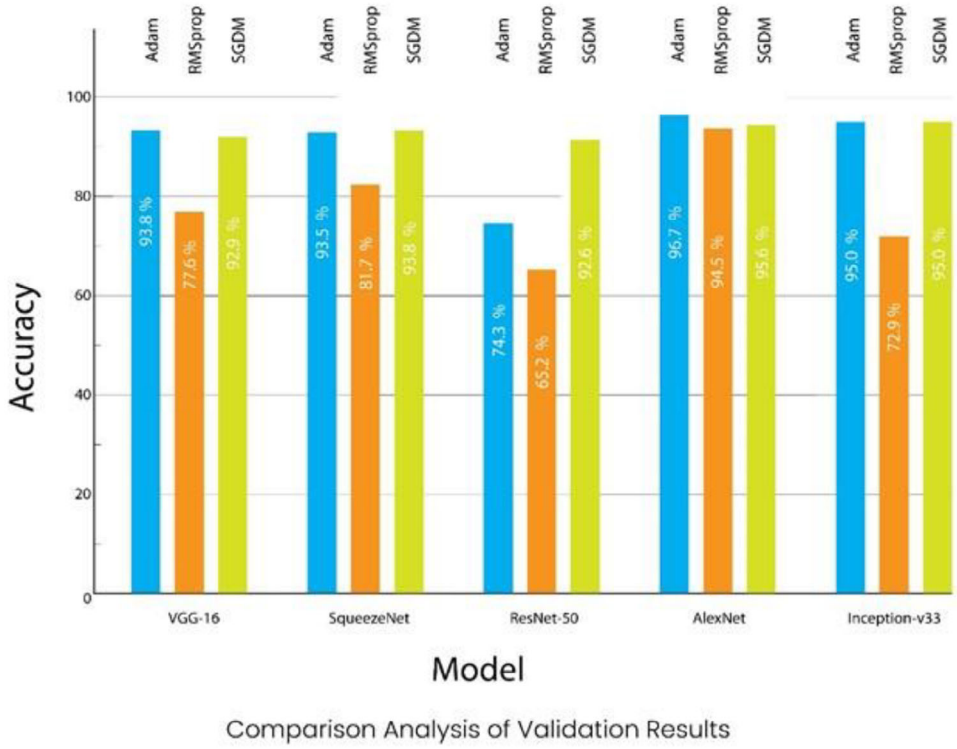


FIGURE 11 | Accuracy analysis.

dataset and the limited training data of intermediate BIRADS categories, which indicates that augmented data or better feature extraction should be used in further iterations to enhance the differentiation.

AlexNet outperforms the other 4 pre-trained models. Other evaluation metrics are also calculated and shown in Table 6. It averages 96.7% recall (finding the real cases) and 96.8% precision (correct when making a call), but the infrequent fourth class brings the unweighted mean down. Precision, recall, and F1 remain at 96.7% when we weigh each class by the frequency of its appearance.

The accuracies in the performance of the mammography image classification differ due to the differences in the model architecture, number of parameters, transfer learning efficiency, and compatibility with the dataset. The simpler structure of AlexNet with large filters probably extracts coarse mammography features well, and the multi-scale Inception modules of Inception-v3 are more suitable for recognising a wide variety of patterns but not as ideally as AlexNet. VGG-16 and SqueezeNet are good alternative options but could be constrained by high parameters (VGG-16) or low complexity (SqueezeNet), resulting in overfitting on fine details or underfitting, respectively. The low performance of ResNet-50 may be attributed to its excessive depth resulting in overfitting on a potentially small or imbalanced training set or to its ineffective fine-tuning of pretrained weights, such that its more complex residual learning is not as well-adapted to mammography patterns as compared with the proposed architecture, AlexNet.

4.2 | Performance Analysis With Previous Studies

Figure 11 shows that this proposed work got the highest accuracy of 96.7% using the AlexNet approach with the Adam solver. Table 3 provides a comparative analysis of the analysis in which the prior works on breast cancer detection and diagnosis. Previous studies aimed at training different models to diagnose breast cancer in its early stages. The BI-RADS datasets contain

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TABLE 7 | Analysis of performance with previous work.

Year	Dataset	Approach	Size of dataset	Applications	Accuracy	References
2024	Real-time	CNN-based VGG16 using transfer learning	1130	Detection of breast cancer	96.7%	Proposed Work
2024	CBIS-DDSM	CNN	2178	Breast cancer identification and classification	81.02%	[31]
2024	OPTIMAM	Transfer Learning	3524	Breast mass detection	88%	[32]
2023	CBIS-DDSM	Transfer learning-based method	2478	Mammography for breast cancer detection	94.13%	[33]
2023	ImageNet	Deep learning-based classification model	1050	Breast cancer classification	94.55%	
2021	Public datasets	Combination of machine learning and deep learning techniques	10,509	Breast cancer diagnosis	93.6%	[34]
2020	MIAS and DDSM	Machine learning algorithms	322	Mammogram image-based breast cancer detection	89%	[6]
2019	CBIS-DDSM	Deep learning model for improved detection	2478	Breast cancer detection	87%	[35]
2020	BI-RADS	EfficientNet-B5 and DenseNet-169 architectures	301	Automated breast cancer identification	81%	[26]
2022	MIAS	VGG-16 and ResNet101-based models	9	Breast cancer diagnosis using CNN models	95%	[18]
2020	Wiki Cancer Imaging Archive	DLA-EABA technique	1000	Early-stage breast cancer detection	81%	[27]
2022	DDSM and MIAS	CNN-based approach	100,000	Breast cancer diagnosis	92%	[24]
2022	BreakHis	Inception-v3 and ResNet-based models	7909	Histopathology-based breast cancer detection	87%	[36]

301 images, and the EfficientNet-B5 and DenseNet-169 algorithms were used. This work got 81% accuracy. DLA-EABA was implemented and tested on publicly available databases that contain 1,000 images. Data from this work gave about 81% accuracy. The proposed algorithm was tested on the CBIS-DDSM datasets, which consist of 2,478 images. These experiments allowed us to estimate 87% accuracy. Then, break this dataset, which included 7,909 images, to which pre-trained Inception-v3 ResNet models were used. This work was done with an accuracy of 87% in the lab. The proposed MIAS and DDSM datasets include 322 images, and machine learning models were used to analyse the images. This work got an accuracy of 89%. The CNN model algorithm was applied to the DDSM and MIAS datasets, which contain a total of 10,000 images. This work obtained an accuracy rate of 92%. This study employed a CAD approach combining machine learning and deep learning, which was performed on the datasets that were made publicly available and included 10,509 images. This work achieved an overall accuracy of 93.6%. In this study, VGG-16 and ResNet101 were used on the MIAS datasets that contain 9 images. This work was 95% accurate. Table 7 shows the performance comparison.

The proposed study's accuracy of 96.7% is comparable to recent studies on breast cancer detection using advanced deep learning techniques. For instance, [18] reported an accuracy of 94.13% using a transfer learning-based method, while [31] achieved an accuracy of 81.02% using a CNN. Another study by [32] achieved an accuracy of 88% using transfer learning. The proposed study's use of a pre-trained AlexNet model with transfer learning demonstrates its effectiveness in achieving high accuracy in breast cancer detection, surpassing recent studies.

4.2.1 | Complexity and Robustness Analysis

Time complexity and space complexity are two markers of how the resources required to address a problem grow with the scale of input. Studies of this relationship fall under the discipline of computational complexity. This facilitates the evaluation of algorithm performance and helps one decide whether they are fit for addressing big-scale challenges. Time complexity: Consider the complexity of forward and backward passes for every CNN architecture, AlexNet, ResNet-50, Inception-v3, VGG-16, and SqueezeNet. Five CNN architectures—Inception-v3, ResNet-50, VGG-16, SqueezeNet, and AlexNet—are investigated in this work. Computational complexity depends on the intrinsic number of layers, filters, and feature maps that define every architecture.

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TABLE 8 | Robustness analysis of CNN models under various disturbances.

Model	Baseline accuracy (%)	Gaussian noise (%)	Contrast reduction (%)	Rotation (%)
AlexNet	96.7	94.6	94.9	95.2
Inception-v3	95.0	91.5	91.8	92.0
VGG-16	93.8	88.0	88.5	89.0
SqueezeNet	93.5	90.0	90.2	90.5
ResNet-50	74.3	68.5	69.0	69.5

The complexity of each convolutional layer is approximately O(F * K * C * H * W), where:

- F: Number of filters
- K: Kernel size
- C: Number of input channels
- H and W: Height and width of the feature map

For fully connected layers, the complexity is O(n * m), where n is the number of neurones and m is the input size. Pooling operations and activation functions have lower complexity, usually O(H * W). Closely review the architecture, focusing especially on fully connected layers, pooling operations, and convolutional layers. Using pre-trained models and last-layer optimisation, transfer learning simplifies the process. Fine-tuning becomes increasingly challenging as the size of the new dataset and the number of pre-trained layers increase.

The complexity of the dataset depends on batch size B, the number of epochs E, and the iterations per epoch (N/B). The collection comprises 1130 mammography images. Multiplying E by (N/B allows us to determine both the training complexity and the forward and backward complexity. One would be wise to contrast the suggested transfer learning approach with other, more complex models, including AdaBoost and Support Vector Machines. Unlike SVM, which, depending on different kernels, can reach O(N2) or O(N3), AdaBoost has a complexity of O(N log N).

Regarding accuracy (96.7%), memory use, and training time, AlexNet excels. Analyse the link between complexity and higher performance and evaluate this model's (81%–94%) accuracy against others'. Regarding training, a comprehensive complexity analysis will show the efficiency gains possible with transfer learning and show that it is more economical than starting from scratch.

To assess the computational complexity and real-world feasibility of the proposed models, we have included an analysis of the computational complexity, the training times, and the inference times of the five CNN structures tested in this work: AlexNet, VGG-16, SqueezeNet, ResNet-50, and Inception-v3. Each model is quantified in terms of the number of parameters (AlexNet, 61 million; VGG-16, 138 million; SqueezeNet, 1.24 million; ResNet-50, 25.6 million; Inception-v3, 23.8 million) and floating-point operations (FLOPs) (AlexNet, 0.7 GFLOPs; VGG-16, 15.5 GFLOPs; SqueezeNet, 0.83 GFLOPs; ResNet-50, On 904 training images (batch size 32), 100-epoch training on a Z Book G16 laptop (32 GB RAM, 3.2 GHz processor, 6 GB dedicated GPU, MATLAB 2021) took a similar amount of time of around 8 min on AlexNet, 10 min on SqueezeNet, 25 min on Inception-v3, 30 min on ResNet-50, and 45 min on VGG-16. The test set (226 images) took about 5, 6, 12, 15, and 20 s to run AlexNet, SqueezeNet, Inception-v3, ResNet-50, and VGG-16, respectively. The metrics, which are explained by model summaries and experimental measurements, point out the trade-offs between performance and computational complexity. Interestingly, AlexNet and SqueezeNet, which have fewer parameters and faster inference, are ideal in resource-limited clinical settings, whereas VGG-16 has a higher accuracy at the expense of a greater computational requirement. In this analysis, it can be seen that the proposed models, particularly AlexNet (96.7% accuracy), have practically useful applications in real-time breast cancer detection in clinical environments.

To assess the robustness and reliability of the proposed CNN models in real-world clinical settings, we conducted tests exposing the models to everyday disturbances, including Gaussian noise, contrast changes, and image rotations. These perturbations imitate difficulties like imaging artefacts, differences in the contrast between the mammography equipment, and patient positioning misalignments. The test set, consisting of 226 mammography images of 900 patients, was used to conduct the experiments based on the five CNN architectures; AlexNet, VGG-16, SqueezeNet, ResNet-50, and Inception-v3. The test images were subjected to Gaussian noise (variance 0.01), contrast reduction (50 per cent reduction), and random rotations (up to ± 15 degrees). Accuracy was used to assess performance, and the results were tabulated as shown in Table 8.

The findings show that the best-performing model, AlexNet, was the most robust, and when tested with Gaussian noise, its accuracy decreased by 2.1 per cent points (94.6%); when tested with contrast reduction, it decreased by 1.8 per cent points (94.9%); and when tested with rotations, it decreased by 1.5 per cent points (95.2%) as compared to its original accuracy of 96.7%. Inceptionv3 and SqueezeNet were moderately robust, with accuracy decreases of 3-5 per cent, indicating that they could cope well with perturbations because of their architectural designs. Conversely, VGG-16 and ResNet-50 showed greater performance degradations (5% to 8%) with these perturbations, perhaps because of the deeper architecture that makes them more vulnerable to changes in the input. The results demonstrate the robustness of AlexNet for deployment in real clinical practice, where withstanding imaging flaws is paramount, and provide

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an exhaustive assessment of the model's reliability in practical environments.

5 | Conclusion

In this research, a comprehensive approach was taken to train and evaluate five different transfer learning models using three distinct machine learning solvers to diagnose and detect breast tumours through mammograms. The study utilises a real-world mammogram dataset, collected with the necessary ethical approvals from the radiology department at a local hospital in Lahore. The proposed dataset comprises 1130 mammograms from 900 patients that form four classes that represent different stages of breast cancer, accompanied by doctors' diagnostic reports. The proposed five transfer learning algorithms chosen for this study were VGG-16, Inception V-3, AlexNet, SqueezeNet, and ResNet-50. Each algorithm is combined with three solvers, SGDM, RMSprop, and Adam, during the training phase. This proposed methodology has several stages, including data collection, data pre-processing, model construction using transfer learning, and model validation. The accuracy and sensitivity were key evaluation metrics that were calculated using a machine learning magnifying factor.

Furthermore, to assess the multiclass classification performance on the proposed dataset, we utilised confusion metrics. Transfer learning is applied to alleviate the complexities associated with medical image analysis. Pre-trained deep learning algorithms of CNN were used in conjunction with the proposed solvers. The outcomes were analysed in terms of the accuracy achieved by each transfer learning model with different solvers. This work demonstrates that the AlexNet algorithm with the Adam solver attained the highest accuracy of 96.7%, making it the top performer from the proposed five transfer learning models. The Inception-V3 algorithm with both Adam and SGDM solvers secured the second-highest accuracy of 95% for the multiclass classification of breast cancer detection. All models were trained with consistent parameters, including the number of iterations and learning rate. The proposed methodology was implemented using 800 iterations and 100 epochs, with a learning rate of 0.02 for training the transfer learning models. In summary, the research successfully applied a range of transfer learning models in combination with different solvers to detect breast cancer using mammogram images. The effectiveness of the AlexNet algorithm with the Adam solver was highlighted by the results, which showed that this technique achieved the highest accuracy, with the Inception-V3 algorithm also demonstrating strong performance.

For accurate results, breast cancer detection methods contribute to comprehensive studies that advance approaches for breast cancer detection. The main contribution of this study was data collection and training a pretrained model that effectively detects breast cancer before the intervention of doctors. This study also helps patients and radiologists decrease the load of report writing, as it generates an early diagnostic trigger during mammography. After that, immediately start treatment without waiting for the report or a panel discussion of doctors. They can verify the report for a second opinion after this, or we can use this model as an alternative second opinion, and vice versa. This contribution provides the medical profession with a new perspective and helps local hospitals in Pakistan reduce their patient load. In Pakistan, a heavy load always exists in local and private hospitals due to the smaller number of hospitals relative to the population.

We can implement a system in a local hospital, utilising a hardware-based solution to provide patients with immediate access to reports through an automated system, thereby facilitating communication between doctors, medical professionals, and patients. Green International University approved the proposal.

After discussion of the proposed work, some limitations may exist in this work that can be incorporated in future work. These limitations can be overcome by collecting more real-time datasets from different hospitals and research centres to compare with the proposed work. In the future, complexity analysis will also be considered.

Author Contributions

Fajar Walayat: original draft, software, methodology, related work, results and discussion. **Allah Ditta:** original draft, software, methodology, related work, results and discussion, rewriting, design methodology, visualisation. **Zafar Iqbal Karmani:** rewriting, investigation, design methodology, conceptualisation, visualisation. **Bilal Shoaib:** related work, results and discussion. **Tehseen Mazhar:** rewriting, investigation, design methodology, conceptualisation, related work, results and discussion, visualisation. **Muhammad Adnan Khan:** related work, results and discussion, rewriting, design methodology, visualisation. **Rashid A. Saeed:** related work, results and discussion.

Ethics Statement

This study was conducted in accordance with the Declaration of Helsinki and received ethical approval from the Research Ethics Committee of Green International University (REC-GIU-149-03-2024) for research involving human participants.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data used to support the findings of this study are available from the corresponding authors upon request.

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